

Lecture 9: Design Recipe

Introduction to Computer Science — BSCS, 1st Semester

1. What is a Design Recipe?

A Design Recipe is a systematic, step-by-step method for solving programming problems and developing correct programs.

Instead of immediately writing code, a programmer first understands the problem, identifies the required inputs and outputs, plans a solution, and then writes and tests the code.

Think of it like constructing a house: you create a blueprint before laying bricks.

Why do we use a Design Recipe?

- To understand problems clearly.
- To reduce programming errors.
- To organize our logic.
- To make programs easier to test and maintain.
- To solve complex problems systematically.

2. The Main Steps of a Design Recipe

We'll use a simple example: calculating the area of a rectangle.

1. Understand the problem

First, identify exactly what the problem is asking.

Example: We need to calculate the area of a rectangle.

2. Identify inputs

Determine what information the program needs.

For our example:

1. Length
2. Width

3. Identify outputs

Determine what result the program should produce.

For our example:

- Area of the rectangle

Formula: $A = \text{length} \times \text{width}$

4. Develop an algorithm

Write the solution as logical steps before coding.

1. Get the length.
2. Get the width.
3. Multiply length by width.
4. Display the area.

Write the program

Translate the algorithm into a programming language.

length = 10

width = 5

*area = length * width*

5. `print(area)`

6. Test the program

Check whether the program gives the expected output.

For length = 10 and width = 5:

$10 \times 5 = 50$ $10 \times 5 = 50$

Expected output: 50

7. Review and improve

Check for errors and consider unusual inputs, such as zero or negative dimensions. Add validation if the problem requires positive dimensions.

3. Design Recipe vs. Algorithm

These two concepts are related, but they are not identical.

Design Recipe	Algorithm
A complete method for designing a program	A sequence of steps to solve a problem
Includes understanding, planning, coding, and testing	Focuses on the actual solution procedure
Helps organize the entire development process	Helps define the program's logic

Remember: An algorithm is one important part of a Design Recipe.

4. A More Formal Design Recipe

In some introductory computer science courses, particularly those influenced by *How to Design Programs*, the term Design Recipe refers to a more specific method for designing functions.

It often includes:

- Signature: What inputs the function accepts and what output it produces.
- Purpose statement: What the function is supposed to do.
- Examples: Sample inputs and expected outputs.
- Template: A structural starting point based on the data.

- Function definition: The actual implementation.
- Tests: Checks that confirm the function behaves as expected.

For example, using Python:

```
# Signature: area(length, width) -> number
```

```
# Purpose:
```

```
# Calculate the area of a rectangle.
```

```
# Example:
```

```
# area(10, 5) = 50
```

```
def area(length, width):
    return length * width
```

```
# Test:
```

```
print(area(10, 5) == 50)
```

Expected output:

True

The formal recipe varies by course and programming language. Your teacher's exact slide sequence takes priority if it differs from this general approach.

5. Why is Testing Important?

A program can run without crashing and still produce an incorrect answer.

For example, imagine accidentally writing:

```
def area(length, width):
    return length + width
```

For inputs **10** and **5**, this gives **15** instead of **50**.

Testing against known examples reveals the mistake.

You should consider:

- Normal inputs.
- Boundary cases.
- Invalid inputs, when relevant.